

GLADYS WEST

American mathematician
1930–present



Modern devices
rely on the Global
Positioning System
(GPS), which was
made possible by
Gladys's work.

Computers can speed up difficult calculations involving huge amounts of data, but only if someone has told them what to do! In the 1960s, the US Navy hired brilliant mathematicians, including Gladys West, to program some of the very first computers.

Gladys wrote software that told computers how to crunch the numbers and to help us understand all kinds of different things, including the strange paths of Neptune and Pluto around the sun. But her most famous project focused on our own planet. The US Navy wanted to put satellites into orbit around the Earth to help ships find their way at sea. They would need to know exactly how far from the ground the satellites were.

GPS works by comparing the distance from a receiver, such as a smartphone, to three or four different satellites, to pinpoint where the receiver is on the Earth's surface.

